PRACTICAL COURSE WORKBOOK

Algorithmic Bias & Fairness

Evaluate technology decisions with evidence and accountability

LEVEL	Intermediate
GUIDED TIME	5 hours across 12 lessons
PROJECT	a documented risk and impact assessment
SKILLS	Bias, Fairness, Audit, ML Ethics, Disparate Impact, Critical evaluation
REVIEWED	2026-08-25

WHAT YOU WILL BE ABLE TO DO

- Use Bias appropriately in a realistic, bounded task.
- Use Fairness appropriately in a realistic, bounded task.
- Use Audit appropriately in a realistic, bounded task.
- Use ML Ethics appropriately in a realistic, bounded task.

WHO THIS IS FOR

Product, policy and technical learners evaluating responsible AI.

BEFORE YOU BEGIN

- No specialist background required

Use this guide beside the interactive lessons. Keep inputs, decisions, tests, limitations and the next improvement.

COURSE MAP

From foundation to finished work

Algorithmic Bias & Fairness is a structured, practical course covering Bias, Fairness, Audit, ML Ethics, Disparate Impact. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 Ethical foundations	1. Stakeholders with Bias 2. Harms with Fairness 3. Rights with Audit
02 Evidence	1. Bias with ML Ethics 2. Measurement with Disparate Impact 3. Impact with Bias
03 Governance	1. Accountability with Fairness 2. Documentation with Audit 3. Oversight with ML Ethics
04 Practice	1. Risk review with Disparate Impact 2. Participation with Bias 3. Monitoring with Fairness

HOW TO STUDY EACH LESSON

1. Read the objective and identify the decision it supports.
2. Reproduce the worked example with the stated input.
3. Test one normal case and one boundary or failure case.
4. Complete the knowledge check without guessing.
5. Save a short evidence note: result, limitation and next action.

CAPSTONE PROJECT

Produce a reviewable Algorithmic Bias & Fairness project using Bias, Fairness, Audit.

DELIVERABLE	A working Algorithmic Bias & Fairness artefact plus an evidence-based self-review.
TOOLS	A suitable Bias environment, A plain-text decision log, Test data or realistic sample material

Production plan

1. Define the intended user, outcome and constraints.
2. Create the smallest complete result using Bias.
3. Test one normal case, one boundary case and one failure response.
4. Revise the work from the evidence and preserve before-and-after results.
5. Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

Evidence log

INPUT	Original brief, data, requirements or test material.
DECISIONS	Chosen method and one reasonable alternative.
TESTS	Normal, boundary and failure results.
LIMITATION	What the result does not establish.
REVISION	One evidence-led improvement.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

Authoritative references

Recommendation on the Ethics of AI

UNESCO - reviewed 2026-08-21
https://www.unesco.org/en/artificial-intelligence/recommendation-ethics

AI Principles

OECD - reviewed 2026-08-21
https://www.oecd.org/en/topics/ai-principles.html

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Bias scope.